

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CS-5002-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Operating System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the evolution of operating system in detail.
b) What are the various services provided by the operating system?
2. a) Compare the following:
i) Spooling and buffering
ii) Hard and soft real time system
b) Differentiate between:
i) System calls and system boot
ii) Multitasking and Multiprogramming
3. a) Explain FcFs and Look disk scheduling algorithm by one example.
b) Explain I/O buffering.
4. a) Discuss various file allocation and access methods. Compare their advantages and disadvantages.
b) What is Thread? What resources are used when a thread is created?

CS-5002-CBGS

PTO

[2]

5.
 - a) Write a semaphore solution for dining philosopher's problem.
 - b) Explain file system used in Linux and Windows.
6.
 - a) What is a deadlock? What methods are used for handling deadlock? How can we get recovery from deadlock?
 - b) What is process? Explain Process Control Block (PCB) with diagram.
7.
 - a) Explain paging and segmentation. How are they helpful in removing fragmentation?
 - b) Discuss the design issues of distributed operating system.
8. Write short notes:
 - a) Distributed Shared Memory (DSM)
 - b) Concurrent Programming
 - c) Parallel Operating System
 - d) Remote Procedure Call (RPC)
